

Technical Progress Report

Project name:	Biped robot		
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1. Executive Summary

This report summarizes the technical progress of the biped robot project in this week. The team has fully completed the mechanical assembly and electrical connection of the biped robot, and verified the normal operation of the camera vision module and the external virtual control website. During the initial walking test, it was found that a single foot cannot support the weight of the whole robot, resulting in unstable standing and failure to complete continuous walking. In addition, the heavy battery brings extra load and the battery fixing scheme is not confirmed yet. The team plans to redesign the foot structure into an I-shape to improve support area and stability, and develop a reliable battery fixing method to ensure stable standing and smooth walking in the next stage.

2.Objectives & Scope

- Complete the overall mechanical assembly and electrical connection of the biped robot.
- Test the initial standing and first step motion of the robot.
- Verify the vision recognition module and remote control website function.
- Identify mechanical stability problems including foot support and battery placement.
- Propose optimization plans for foot structure and battery fixing.

3. Technical Work Completed

3.1 Mechanical Assembly & Integration

Completed the full assembly of the biped robot structure, including torso, legs, servos, brackets and connectors. Installed and fixed all servos, connecting rods and joint components correctly. Completed wiring arrangement and fixed the main controller and driver modules.

3.2 Electrical System & Debugging

Finished power supply connection, signal wiring and servo parameter testing. Verified normal signal communication and basic action response. Confirmed that the motor and servo system can be driven normally.

3.3 Vision Module & Virtual Website

Completed the debugging of the ESP32-CAM vision module, realizing stable color recognition. Verified the external virtual control website, which can remotely adjust servo angles.

Confirmed the coordination between vision system and robot control system.

3.4 Initial Motion Test

Carried out the first standing and first step test of the robot.

Recorded the support force, balance state and motion stability.

4. Observations & Findings

- The overall assembly of the robot is completed successfully, and all electrical systems work normally.
- When the robot tries to take the first step, one foot cannot support the total weight, causing obvious tilt and instability.
- The current foot structure has small contact area and weak support capacity, which cannot meet the balance requirement.
- The battery is relatively heavy, increasing the overall load and affecting the standing stability.
- There is no reasonable fixing position for the battery, which may lead to shaking or falling during movement.
- The vision recognition and remote website functions remain stable and reliable.

5.Challenges & Roadblocks

- Single foot support capacity is insufficient, unable to bear the robot's weight during walking.
- Original foot structure has small stress area and poor stability.
- Heavy weight of the battery increases the load on legs and affects balance.
- Lack of a safe, firm and reasonable battery fixing scheme.
- The walking test cannot continue until the foot structure and battery layout are optimized.

6. Next Steps & Recommendations

- Redesign the robot foot into an I-shape structure to increase support area and improve stability.
- Print and install the new I-shape feet, then re-test the standing and single-leg support.
- Design and implement a reliable battery fixing mechanism to secure the battery firmly.
- Adjust the battery position to lower the center of gravity and improve balance performance.
- Re-test the initial standing and first step motion after structural modification.
- Continue optimizing the vision recognition algorithm and remote control website.

7. Conclusion

This week, the team successfully completed the full assembly and initial debugging of the biped robot. The electrical system, vision module and remote control website work well. However, the robot cannot stand stably on one foot during the first step test due to insufficient support capacity of the foot structure. Meanwhile, the heavy battery brings extra load and its fixing method remains a problem. The project will focus on redesigning the I-shape foot structure and determining the battery fixing scheme to improve support and stability. After optimization, the team will conduct walking tests again to ensure the robot can realize stable standing and continuous motion.